VPAT 4XXX/6XXX

Computational Comparative Pathology

Artificial Intelligence as a Force-Multiplier for the Diagnostic Pathology Service

Proposed Course Syllabus

Offered by: Department of Pathology, College of Veterinary Medicine, University of Georgia
Cross-listing: UGA Institute for Artificial Intelligence
Open to: Undergraduates (4XXX) and graduate students (6XXX): CBS pathology and diagnostic-imaging students, DVM/PhD trainees, residents, AI-program and AI-certificate students, and advanced undergraduates in the biological and animal sciences, with instructor or advisor approval
Levels: 4XXX (undergraduate)/6XXX (graduate), with differentiated requirements (see Grading)
Credit hours: 3
Format: Weekly 3-hour seminar interleaving discussion, hands-on lab work, and a standing data clinic for your own data · 15 weeks
Course materials: lab notebooks, recipe cards, and curated data subsets, provided via the course repository
Instructor: Bin Duan, PhD. Research focus: AI and Computational Tools for Biomedical Imaging.
Contact / office hours: email TBD; weekly office hours TBD, and by appointment
Meeting time / location: TBD
TA: TBD
TA Contact / office hours: TBD
Exam: No
Prerequisites: None

1. Course Description

Veterinary diagnostic services are understaffed and the caseload keeps climbing, much of it repetitive counting and measuring. This course shows you how artificial intelligence (AI) can take those routine tasks off your plate, so your time goes where your judgment matters. It is **comparative** by design: its core question is why a tool that works on one scanner, lab, or species can quietly fail on another. That gap, **domain shift**, is the field's central puzzle and the bridge between veterinary and human medicine under the Precision One Health Initiative.

You do **not** need to code or do math beyond what you already use. Each week pairs a pathology idea you know with one new computational idea, and in the labs you run and tweak small, clearly labeled cells of provided notebooks rather than build software. One case study threads the term, **quantifying tumors from H&E**: you outline nuclei and regions, then turn those outlines into counts and grades, anchored on mitotic counting, where experts disagree and the count changes the grade. Labs use free, open tools (QuPath, public datasets, browser notebooks, and the **OnePath** copilot in a few later weeks). You finish with a **project on your own data**, a short plain-language brief presented to the class. The aim is concrete: by the end, you can take one of these tools to your own slides and get a result you can defend.

2. Learning Objectives

By the end of this course, you will be able to:

1. **Describe** how a glass slide becomes digital, walk the histopathology workflow, and name where AI can help in it.
2. **Explain**, in plain language, how a model learns and is tested, and what overfitting, classification, detection, and segmentation mean.
3. **Run** a provided pipeline (normalize, tile, outline or detect) on slide data, including your own, and interpret its output for grading.
4. **Judge** whether a model can be trusted: how it fails (domain shift across scanners, labs, and species; noisy labels) and which honest measures (agreement, reproducibility, sensitivity and specificity) beat a single accuracy number.
5. **Carry out and present** a project on your own data, written up as a short plain-language brief.

3. Prerequisites, and Using These Tools on Your Own Data

Prerequisites: none. No computer science, programming, statistics, or calculus. This course is for biomedical scientists and clinicians new to AI: every idea starts from zero, motivated by a pathology problem first. In the labs you run and tweak small, clearly marked cells of provided notebooks (the kind of editing you would do to a spreadsheet formula), never writing software from scratch. All you need is a laptop with a web browser; heavy computation runs in the cloud (Google Colab or our JupyterHub) or in OnePath, which needs no setup.

Your own problem, from week one. The course is built so you can point these tools at your own slides and data, not just the class demos. Five things make that realistic even for non-coders:

- **Bring your own data early.** From Week 2 on, most labs have two halves: run the steps on the shared demo, then run the same steps on a few of your own images. You work on your real problem all term, not only at the end.
- **Recipe cards, not programming.** Every tool comes with a one-page recipe: what you need, the single button or cell to change, what the output means, and the common errors. You copy a recipe and change one thing, usually a folder name.
- **Point-and-click first.** Wherever possible you drive tools by clicking (QuPath, OnePath); notebooks are the fallback, and they come pre-filled so you only edit clearly marked cells.
- **Use an AI assistant to adapt the code.** You will practice asking an AI assistant to adjust a notebook for your data ("load my folder of images instead of the sample"), and, just as important, how to check it did the right thing.
- **A weekly data clinic.** Part of each session is hands-on time to get your own data unstuck, with the instructor and your classmates. Bring whatever is breaking.

By the final project, running a tool on a fresh set of your own cases will feel routine.

4. Weekly Schedule

Each week pairs a pathology reason (**Why it matters**) with one new computational idea (**What's new**), then puts it in your hands in the **Lab**. Our running case study, **quantifying tumors from H&E**, starts in Week 7: you outline nuclei and regions, then turn those outlines into the counts and grades a report depends on. Notes marked *Optional deep-dive* are extra material for the curious; nothing in a deep-dive is required or graded.

Schedule at a glance

The 15 weekly sessions meet Tuesdays, Fall 2026 (*tentative*). Thanksgiving week has no class. Full week descriptions follow the table.

Wk	Date	Topic	Project milestone
1	Aug 18	Why computational pathology, and where it helps	—
2	Aug 25	Whole-slide imaging: from glass to pixels	first "your data" step
3	Sep 1	The histopathology workflow, and where AI inserts	—
4	Sep 8	Stain and color normalization	—
5	Sep 15	How a model learns, without the magic	choose question + data
6	Sep 22	Deep learning, demystified	**proposal due**
7	Sep 29	Outlining nuclei and regions (case-study kickoff)	—
8	Oct 6	From outlines to numbers: counts, cellularity, grade	—
9	Oct 13	Domain shift: scanners, labs, species	—
10	Oct 20	Label noise and the trouble with ground truth	—
11	Oct 27	Registration: lining up sections, stains, species	method finalized
12	Nov 3	Image + report: AI as decision support	—
13	Nov 10	Human-in-the-loop and putting tools to use	—
14	Nov 17	Evaluation pathologists trust	**draft for peer review**
—	Nov 24	*No class — Thanksgiving break*	—
15	Dec 1	Project showcase (invited panel)	**final brief due**

Week 1 — Why Computational Pathology, and Where It Helps

- **Why it matters:** The understaffed diagnostic service: caseload, turnaround time, burnout, and the parts of sign-out that are repetitive and quantitative.
- **What's new:** What AI *is* and *is not*, in plain terms: pattern recognition versus "intelligence," where it helps, and where it fails.
- **Lab:** Install and tour **QuPath**; open a whole-slide image and move around it, then look at one finished AI result so you can see where the term is headed. No code.
- **Reading:** Zuraw & Aeffner, "Whole-slide imaging, tissue image analysis, and artificial intelligence in veterinary pathology: An updated introduction and review," *Vet Pathol* (2022). Op-

tional: Niazi, Parwani & Gurcan, "Digital pathology and artificial intelligence," *Lancet Oncol* (2019).

Week 2 — Whole-Slide Imaging: From Glass to Pixels

- **Why it matters:** The scanned slide as a clinical object: what is gained and lost versus the microscope, plus magnification, focus, and scan artifacts.
- **What's new:** An image is just a grid of numbers; resolution and magnification; why a slide is stored as a zoom **pyramid** and chopped into **tiles** (you cannot open a 100,000 $\times$ 100,000-pixel slide all at once).
- **Lab:** Open a slide, cut it into tiles at a few magnifications, and look at the pyramid. Brought a slide of your own? Do the same with it; that is your first "your data" step.
- **Reading:** Aeffner et al., "Introduction to Digital Image Analysis in Whole-Slide Imaging: A White Paper from the Digital Pathology Association," *J Pathol Inform* (2019).

Week 3 — The Histopathology Workflow and Where AI Inserts

- **Why it matters:** Gross $\rightarrow$ trim $\rightarrow$ cassette $\rightarrow$ process $\rightarrow$ embed $\rightarrow$ section $\rightarrow$ H&E $\rightarrow$ sign-out, and the variation or artifact that creeps in at each step.
- **What's new:** Where the data comes from and why it matters: what counts as a "sample," how early choices ripple downstream, and why "garbage in, garbage out" is literal here.
- **Lab:** Trace a real case end to end; mark tissue regions and artifacts in QuPath.
- **Reading:** Bertram & Klopfleisch, "The Pathologist 2.0: An Update on Digital Pathology in Veterinary Medicine," *Vet Pathol* (2017).

Week 4 — Stain and Color Normalization

- **Why it matters:** H&E looks different across labs, days, and stainers: the batch effects you already recognize at the scope.
- **What's new:** Why color differences trip up a model, and how we even out color so the tissue, not the stain, drives the result (we will look at this by eye, not by equation).
- **Lab:** Even out the color across slides from two labs; see the before/after and how it changes a simple classifier.
- **Reading:** Macenko et al., "A method for normalizing histology slides for quantitative analysis," *ISBI* (2009); read for the intuition, not the math.

Week 5 — How a Model Learns, Without the Magic

- **Why it matters:** "Tumor versus not-tumor on this tile" framed as a decision a model can be trained to make.
- **What's new:** How a model learns from examples; the **train / validation / test** split; and **overfitting**, why a model that aces practice can flunk the real exam.
- **Lab:** Train a tile classifier (tumor versus non-tumor) in a provided notebook; overfit it on pur-

pose and watch its test score collapse.

- **Reading:** A short plain-language primer (provided) plus a glossary handout.

Week 6 — Deep Learning, Demystified

- **Why it matters:** How your eye sweeps a slide for patterns, and how a network learns its own patterns the same way.
- **What's new:** Neural networks and "convolution" as pattern detectors, in pictures; and **transfer learning**, reusing a model someone else already trained. No heavy math.
- **Lab:** Run a pretrained network on tiles and adapt it with transfer learning; compare it to Week 5's classifier. Then the term's key independence skill: ask an AI assistant to retarget a provided notebook to your own folder of images, run it, and check the result is right (the expected number of tiles, sensible labels). From here on, this is how you put any lab onto your own data.
- **Reading:** A short, figure-driven explainer (provided). Optional: LeCun, Bengio & Hinton, "Deep learning," *Nature* (2015); skim.

Week 7 — Case Study Kickoff: Outlining Nuclei and Regions

- **Why it matters:** Almost every number in a report starts with a boundary: where the tumor is, where the necrosis is, which cells to count. Drawing those by eye is slow and hard to reproduce, yet it drives grading and margins.
- **What's new:** Two kinds of outlining (separating tissue types versus separating one cell from the next), the U-Net idea in pictures, and a simple score for how well the computer's outline overlaps the one you would draw. A sloppy outline quietly corrupts every count built on it.
- **Lab:** Outline nuclei and regions on **PanNuke** or **MoNuSeg**; overlay the result on H&E in QuPath; fix an outline by hand and watch the count change.
- **Reading:** Ronneberger, Fischer & Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," *MICCAI* (2015); read the figures and the intuition. Optional: Graham et al., "Hover-Net: Simultaneous segmentation and classification of nuclei," *MIA* (2019).

Week 8 — From Outlines to Numbers: Counts, Cellularity, and Grade

- **Why it matters:** Outlines only help once they become numbers you act on: nuclear count and density, mitotic figures per area, necrosis fraction, margin distance. This is where outlining meets grading.
- **What's new:** Turning outlines into measurements; why the measurement, not the pretty overlay, is the deliverable; and finding and labeling specific objects (like mitotic figures) on top of an outline.
- **Lab:** Turn last week's nuclei outlines into counts and density; then flag **mitotic figures** on **MIDOG** and **CCMCT** tiles and compute a count of the kind that feeds a grade.
- **Reading:** Bertram et al., "A large-scale dataset for mitotic figure assessment on whole slide images of canine cutaneous mast cell tumor," *Scientific Data* (2019).

Week 9 — Domain Shift: Scanners, Labs, Species

- **Why it matters:** The same measurement task, on a different scanner, in a different lab, or in a different species. Why a tool that was "validated" can quietly get worse.
- **What's new:** Why models break on data that looks a little different, and the everyday fixes (showing the model more varied examples) that help.
- **Lab:** Take a **provided** mitosis detector, run it on one MIDOG group and then on another (different scanners, different species), and measure how much it drops. Then look at where the same model lands on dog versus human tiles, so you see the shift rather than just read about it. You drive a ready-made detector, so there is nothing to train; the payoff is watching a "validated" model wobble the moment the species changes.
- **Reading:** Aubreville et al., "Mitosis domain generalization in histopathology images: The MIDOG challenge," *Medical Image Analysis* (2023). Optional: Aubreville et al., "Domain generalization across tumor types, laboratories, and species: insights from the 2022 MIDOG challenge," *Medical Image Analysis* (2024).

Week 10 — Label Noise and the Trouble with Ground Truth

- **Why it matters:** Expert labels are not perfectly repeatable, whether it is where to draw a boundary or which figures count as mitoses. A second pathologist, or the same one on another day, may disagree. So what *is* "ground truth"?
- **What's new:** Label noise; why we use a panel of experts; and how shaky labels limit both what a model can learn and what its scores can honestly mean.
- **Lab:** Compare two annotators on the same slides and measure how often they agree (Cohen's / Fleiss' κ); talk through what a model can and cannot claim on top of that.
- **Reading:** Bertram et al., "Are pathologist-defined labels reproducible? Comparison of the TUPAC16 mitotic figure dataset with an alternative set of labels," *MICCAI 2020 Workshops (LABELS)*, Springer LNCS (2020).

Week 11 — Registration: Lining Up Sections, Stains, Species

- **Why it matters:** Lining up serial sections, or matching an H&E to its IHC, so you are reading *the same spot* across slides, the groundwork for any multi-stain comparison.
- **What's new:** Registration: sliding and gently stretching one image until it matches another; when a simple shift is enough and when you need to warp; and how warping can sometimes invent detail that was never there.
- **Lab:** Drive a prepared registration notebook (you change the inputs; it does the alignment): line up serial sections ANHIR-style and align an IHC slide to its H&E, and watch where stretching helps and where it invents detail. You probe a ready-made pipeline rather than build one; the memorable part is catching it fabricate tissue that was never on the slide.
- **Reading:** Borovec et al., "ANHIR: Automatic Non-rigid Histological Image Registration Challenge," *IEEE TMI* (2020).

Week 12 — Image + Report: AI as Decision Support

- **Why it matters:** The **report**, not a heatmap, is what leaves your office. Could a tool read the picture and the words together and help you write a cleaner, more consistent report, without writing it for you?
- **What's new:** Vision-language models: tools that connect what is in an image to words, so a computer can describe a tile or pull up similar described cases. We will also name where they go wrong: making things up, and sounding confident when they should not.
- **Lab:** Use **OnePath's copilot** on a slide and its report together: ask it about the case, and have it tidy a free-text report into clear, structured fields. *Optional deep-dive: the same copilot can also fold in molecular (gene-expression) data and show which biological pathways it leaned on, available as an extra notebook for anyone curious.*
- **Reading:** Lu et al., "A visual-language foundation model for computational pathology" (CONCH), *Nature Medicine* (2024). Optional: Huang et al. (PLIP), *Nature Medicine* (2023); and the newer image foundation models for pathology, e.g. Chen et al., "Towards a general-purpose foundation model for computational pathology" (UNI), *Nature Medicine* (2024).

Week 13 — Human-in-the-Loop and Putting Tools to Use

- **Why it matters:** Annotation workflows, the sign-out review loop, and what it actually takes for a pathologist to **trust** and adopt a tool.
- **What's new:** Keeping a human in charge: how a tool can suggest, you correct, and those corrections make it better over time; plus the things that go wrong once a tool is deployed.
- **Lab:** Use **OnePath's experts-in-the-loop** panel: confirm or flag each call; your corrections are saved and used to improve the tool. We will talk through why this review step is what earns trust.
- **Reading:** Selected deployment / human-in-the-loop perspective (provided); a short, plain overview of how digital-pathology tools are regulated.

Week 14 — Evaluation Pathologists Trust

- **Why it matters:** "95% accurate" can be meaningless at sign-out. What pathologists trust is **agreement** and **reproducibility**: does the tool agree with the expert, and with itself?
- **What's new:** A few honest yardsticks: **sensitivity** and **specificity** (catching disease versus avoiding false alarms) and **agreement with the pathologist**. The right yardstick depends on the clinical question.
- **Lab:** Score your project model two ways, a standard accuracy-style number and agreement-with-the-pathologist, and see how differently they can read. *Optional deep-dive: OnePath's evaluation panel also shows whether a model "knows when it is unsure" and can choose to hold back rather than guess.*
- **Reading:** Aeffner et al., "The Gold Standard Paradox in Digital Image Analysis: Manual Versus Automated Scoring as Ground Truth," *Archives of Pathology & Laboratory Medicine* (2017).

Week 15 — Project Showcase

- **Why it matters:** A result only counts if the people who would use it believe and adopt it. You present to the audience that matters, not just the instructor.
- **What's new:** Pulling it together: placing your project on the themes we have used all term (domain shift, label noise, honest evaluation).
- **Lab / Assessment:** Present your project to the class and an invited panel of pathologists, residents, and PIs, and submit your plain-language brief. The aim is a result they would actually act on.
- **Reading:** Your classmates' projects.

5. Grading

Component	Weight
Term project — proposal (10%) + final brief (30%) + presentation (10%), on your own data	**50%**
Weekly labs — your best 8 of ∼10 submitted notebooks	**30%**
Participation — reading responses, discussion, and the weekly data clinic	**15%**
Peer review of a classmate's project draft	**5%**
Total	**100%**

Basis for grade. Final percentages convert to letter grades on the scale below (rounded to the nearest whole percent). Graduate programs typically require a B or better in elective coursework; confirm the minimum with your program.

A	93–100	C+	77–79
A-	90–92	C	73–76
B+	87–89	C-	70–72
B	83–86	D	60–69
B-	80–82	F	below 60

Your term project. You apply what you have learned to your own slides or data. No data of your own? You can use a guided public dataset matched to your area, or run OnePath on a fresh set of slides and report what you find; either way, no coding is required. By the time you start, you will have run these steps on your own data all term, so the project applies a familiar workflow rather than teaching a new one. We build it up across the term: pick a question and data (Weeks 5–6), choose an approach (Weeks 7–11), plan how you will judge it (Week 14), and present (Week 15). The written deliverable is a short plain-language brief (about four pages) with set sections: clinical question and motivation, data, what you did, an honest read of whether it works and how it could fail (such as domain shift or label noise), limitations, and next steps, with your figures and notebook attached. Write it for a clinical or research reader who does not work in AI. You are graded on honest framing and honest evaluation (does it really work, and how might it fail?), not on chasing the highest accuracy number.

Levels (4XXX undergraduate / 6XXX graduate). The course is dual-listed: all students attend the same sessions and labs and are graded on the same weights. Graduate (6XXX) students complete the term project on their own dissertation or diagnostic data, with a deeper evaluation and one additional paper critique; undergraduate (4XXX) students complete the same project on a provided public dataset at a lighter scope, with the additional critique optional. Expectations scale with level.

Project rubric (the four-page brief, 30% of the grade). The brief is scored on:

Criterion	Weight
Clinical question and motivation — real, specific, and well-scoped	15%
Data and method — appropriate choice, correctly applied, clearly described	25%
Honest evaluation — the right measures, plus an explicit account of how the tool could fail on your data (domain shift, label noise)	30%
Limitations and next steps — candid and specific	15%
Communication — clear to a non-AI reader, figures that support the claims, within ∼4 pages	15%

An A brief poses a real clinical question, applies a method correctly, and is honest about where the result would break; a model that fails gracefully and says so scores higher than a higher-accuracy black box. The proposal (Week 6, 10%) and presentation (Week 15, 10%) are graded on the same priorities at the draft and spoken stages.

Participation (15%) rewards consistent, good-faith engagement: a brief written response to each week's reading (credit / no-credit), active discussion, and bringing your own data to the weekly clinic. **Peer review (5%):** in Week 14 you review one classmate's draft against the rubric above using a short provided form, graded on how useful your feedback is, not on the score you assign.

No exams. Everything is project- and lab-based, in keeping with the program's applied, area-of-emphasis style.

6. Datasets and Tools

Everything below is free and public. We chose datasets that are veterinary, comparative (the same task in animals and humans), or both. Links are starting points; check the access terms when you use them. For class use, labs run on small, pre-tiled subsets of these datasets (curated and provided), so nothing requires downloading a full whole-slide archive or a GPU.

Getting your own data in (a starter recipe).

- Whole-slide files are usually `.svs` (Aperio), `.ndpi` (Hamamatsu), `.mrxs` (3DHistech), or `.tiff`; ordinary photos and exports are `.png` or `.jpg`. QuPath and OpenSlide open the slide formats directly; OnePath and the notebooks take `.png` / `.jpg` tiles.
- To start, drag a slide into QuPath, or drop a folder of `.png` / `.jpg` images into OnePath or the lab notebook. Nothing to convert for png/jpg; for a whole slide, the Week 2 notebook cuts it into tiles for you.

- Stuck loading your data? That is exactly what the weekly data clinic and the AI-assistant recipe (Week 6) are for.

Outlining nuclei and regions (the running method).

- **PanNuke** — pan-cancer nuclei outlining and classification (TIA Centre, University of Warwick; PanNuke is under "Data"). `https://warwick.ac.uk/TIA`
- **MoNuSeg** — multi-organ nuclei outlining. `https://monuseg.grand-challenge.org/`

Mitotic counting and grading (the clinical application, with cross-species data).

- **MIDOG 2021** — cross-scanner mitosis (human breast). `https://midog2021.grand-challenge.org/`
- **MIDOG 2022** — multi-tumor, multi-species (incl. canine) mitosis. `https://midog2022.grand-challenge.org/`
- **MIDOG++** — multi-domain mitotic-figure dataset (7 tumor types incl. canine soft-tissue sarcoma; human + canine), Aubreville et al., *Scientific Data* (2023). `https://github.com/DeepMicroscopy/MIDOGpp`
- **Canine Cutaneous Mast Cell Tumor (CCMCT)** mitosis dataset — Bertram et al., *Scientific Data* (2019). `https://github.com/DeepMicroscopy/MITOS_WSI_CCMCT`
- **Canine Mammary Carcinoma (CMC)** mitosis dataset — DeepMicroscopy. `https://github.com/DeepMicroscopy/MITOS_WSI_CMC`
- **TUPAC16** — Tumor Proliferation Assessment Challenge (mitosis, human). `https://tupac.grand-challenge.org/`

Registration.

- **ANHIR** — automatic non-rigid histological image registration (serial sections, multi-stain). `https://anhir.grand-challenge.org/`

Detection / metastasis (human).

- **CAMELYON16/17** — lymph-node metastasis detection across centers. `https://camelyon17.grand-challenge.org/`

General slide repositories (for your own / area-of-emphasis projects).

- **The Cancer Genome Atlas (TCGA)** WSIs via the GDC Data Portal. `https://portal.gdc.cancer.gov/`
- **The Cancer Imaging Archive (TCIA).** `https://www.cancerimagingarchive.net/`

Tools (all free).

- **QuPath** — free whole-slide viewer and analysis; the everyday tool for opening, annotating, and measuring slides. `https://qupath.github.io/`
- **Google Colab / institutional JupyterHub** — notebooks that run in the cloud (nothing to install).
- Provided notebooks with all the computer-science plumbing hidden behind clearly marked, editable cells.

- **OnePath** — an open, comparative (One Health) pathology copilot featured in the multimodal, human-in-the-loop, and evaluation weeks: read an image alongside its report, keep a human in the loop, and evaluate honestly. It opens on your laptop with no setup, no GPU, and no internet. A research prototype for decision support, not a clinical device.

7. Course Policies

Academic Integrity. UGA has a University Honor Code and Academic Honesty Policy, and all students are responsible for knowing and upholding them. All academic work must meet the standards described in *A Culture of Honesty*; you are responsible for informing yourself of those standards before doing any academic work, and ignorance is not an acceptable defense. The UGA Student Honor Code states: "I will be academically honest in all of my academic work and will not tolerate academic dishonesty of others." If you are unsure whether a particular collaboration or tool use is allowed, ask before you submit.

Using AI tools. This is a course about AI, so using AI tools is part of the work: you are encouraged to use AI coding assistants to adapt notebooks to your data, and OnePath and similar tools are course material. Three rules keep that honest:

1. **Disclose it.** Note where and how you used an AI tool in anything you hand in; a sentence is enough.
2. **You own the output.** You are responsible for checking that anything an AI tool produces is correct. "The model said so" is never a defense, and unverified output that turns out wrong is your error.
3. **Do your own thinking.** AI may help you write code and tidy prose, but the analysis, interpretation, and conclusions in your project must be your own. Generating those wholesale with AI, or without disclosure, is an integrity violation.

Disability and access. UGA is committed to equal access. If you have a disability or condition that may affect your participation, please register with Accessibility and Testing (Clark Howell Hall, 706-542-8719) and share your accommodation letter with me early in the term. Accommodations are not retroactive.

Attendance and participation. Much of the value is hands-on: the labs and the weekly data clinic happen in class, and participation is part of your grade. Regular attendance is expected; if you must miss a session, let me know in advance and I will share materials where possible.

Late work and makeups. Lab notebooks are due before the next session; project milestones (proposal Week 6, draft Week 14, final brief Week 15) have fixed dates. Late work loses 10% per day to a floor of 50%, unless you arrange an extension with me in advance. The best-8-of-10 lab policy already absorbs two missed or weak labs. If a documented illness or emergency causes you to miss a graded deadline, contact me as soon as possible (before the deadline where you can) to arrange a makeup or a shifted milestone.

Mental Health and Wellness Resources.

- If you or someone you know needs assistance, you are encouraged to contact **Student Care**

and Outreach in the Division of Student Affairs at **706-542-7774** or visit sco.uga.edu. They will help you navigate any difficult circumstances you may be facing by connecting you with the appropriate resources or services.

- UGA has several resources for students seeking mental health services or crisis support.
- If you need help managing stress, anxiety, relationships, etc., please visit BeWellUGA for a list of FREE workshops, classes, mentoring, and health coaching led by licensed clinicians and health educators in the University Health Center.
- Additional resources can be accessed through the UGA App.

Resources for Student-Parents. If you or someone you know is in a phase of life that involves parenting (or the expectation of parenting), Student Care and Outreach within the Office of the Dean of Students can provide information and resources (706-542-7774). The student group UP at UGA offers peer support to pregnant and parenting students (upatuga.org).

This syllabus is a general plan. Deviations announced in class (content, schedule, or due dates) may be necessary; students are responsible for changes announced in class even if not present.

Instructor note: before the course runs, confirm the current UGA required statements (academic honesty, Mental Health and Wellness Resources, accessibility, FERPA, non-discrimination) and the public-syllabus posting requirement via the CTL syllabus checklist and syllabus.uga.edu.